

MS-2000 Parts and Materials

Design A, Pan and Tilt Camera Turret. Design locked Sep 16, 2026.

Links checked
Sep 15 to 16, 2026
15 line items to buy

\$191.94

Primary order, 2 suppliers: DFRobot \$154.70 + DigiKey \$37.24. Shipping and tax extra.

~\$226.94

All-in with about \$35.00 of batteries, foam board, rod, and craft supplies from local stores.

\$174.35

If every backup were used instead (Adafruit \$88.65, DigiKey \$73.80, DFRobot \$11.90), not counting the backup laser price.

How to read this

- Every part has a **primary** and a **backup**. Buy the primary. Use the backup only if the primary is out of stock or late.
- Primary uses **two suppliers**: DFRobot (the micro:bit, camera, board, and servos all come from one ecosystem) and DigiKey (the laser and small parts).
- Backups use Adafruit and DigiKey, plus DFRobot for the backup board.
- Every link was opened on the dates above. Prices are what each site showed.
- Amazon is left out on purpose: its listings change sellers and cannot be checked reliably.

Check before and after ordering

- **DigiKey prices marked *** come from DigiKey's listing on Octopart; confirm in the cart.
- **DigiKey stock is thin** on the 4-wire cable (16). Order it with everything else.
- **DFRobot** ships from its US warehouse by UPS (3 to 7 days to Texas). Items not in the US warehouse ship from China in 10 to 17 business days with duties prepaid.
- **On arrival**: check the Xia mi board's external power connector (the pack has a DC 2.1 jack), whether the board powers the micro:bit, and which 5 V port the servos run best on.
- **Laser**: read the Class II label on the module before first power-up.

Brains and control

01 micro:bit V2

Qty 2

One runs the turret, one lives in the mosquito wand. They talk by radio, and the wand one records the data.

PRIMARY	DFRobot	MBT0039	In stock	
	micro:bit V2			\$45.80 2 × \$22.90
BACKUP	Adafruit	4781	In stock	
	BBC micro:bit v2			\$35.90 2 × \$17.95

Board only. Uses micro-USB for code; a cable is almost certainly on hand.

02 Expansion board (turret)

Qty 1

Holds the turret micro:bit. 5 V ports for the servos and speaker, a 5 V HuskyLens port, a relay that switches the laser, and an external power input.

PRIMARY	DFRobot	MBT0042	In stock	
	Xia mi Multi-functional Expansion Board for micro:bit V2			\$29.90
BACKUP	DFRobot	DFR0548	In stock	
	micro:Driver expansion board (8 servo, 4 motor)			\$11.90

Xia mi: HuskyLens port (5 V I2C), 5 V GPIO on P0 P1 P2 P8, relay and 4 motor channels; 5 to 12 V external input, MakeCode extension DFRobot/pxt-DFRobot_xia_mi_Board. Backup board: servos and motor channels confirmed; its I2C port voltage for the HuskyLens is not documented.

Camera and aiming

03 HuskyLens AI camera

Qty 1

Learns the fake mosquito and reports where it is on screen.

PRIMARY	DFRobot SEN0305 In stock Gravity: HuskyLens K210 AI Camera dfrobot.com/product-1922.html	\$34.90
BACKUP	DigiKey 1738-SEN0305-ND 68 in stock* DFRobot SEN0305 (HuskyLens) digkey.com/en/products/detail/dfrobot/SEN0305/10529331	\$54.90

Comes with a 4-pin Gravity cable, two brackets, and M3 screws and nuts for its own mounting. MakeCode extension: DFRobot/pxt-DFRobot_HuskyLens.

04 Micro servo

Qty 3

Pan, tilt, and one spare.

PRIMARY	DFRobot SER0049 In stock 9g 180 degree Clutch Servo (damage-proof) dfrobot.com/product-2120.html	\$14.70 3 × \$4.90
BACKUP	Adafruit 169 In stock Micro Servo, TowerPro SG92R adafruit.com/product/169	\$17.85 3 × \$5.95

Clutch servo slips instead of stripping gears when a hand grabs the turret. 4.8 to 6 V. Price is the 3+ rate (\$5.00 each for 1 or 2).

Laser and hit sensing

05 Class 2 laser module, under 1 mW

Qty 1

The laser. Maker-labeled Class II.

PRIMARY	DigiKey VLM-650-03-LPT-ND 1,372 in stock* Quarton VLM-650-03 LPT, 650 nm, <1 mW, 7 mm digkey.com/en/products/detail/quarton-inc/VLM-650-03-LPT/1010454	\$25.03
BACKUP	DigiKey VLM-650-22 LPT Ships today Quarton VLM-650-22 LPT, 650 nm, 0.5 to 0.9 mW, 4 mm digkey.com/en/products/detail/quarton-inc/VLM-650-22-LPT/8636974	at checkout

Quarton manuals confirm both LPT versions are Class II, under 1 mW. Not sold by Adafruit, DFRobot, or Mouser. Do not substitute a 5 mW hobby laser (Class IIIa).

06 Laser arm switch

Qty 1

Cuts power to the laser only. Must be on before the laser can fire at all.

PRIMARY	DigiKey Adafruit 805 Price not shown Breadboard-friendly SPDT slide switch digkey.com/en/products/detail/adafruit-industries-llc/805/26735538	\$0.95
BACKUP	Adafruit 805 In stock Breadboard-friendly SPDT Slide Switch adafruit.com/product/805	\$0.95

DigiKey lists this part but did not show a price; \$0.95 is Adafruit's price. Confirm in the cart, or add it to an Adafruit order.

07 Light sensor

Qty 3

One inside each mosquito body. Feels the laser dot.

PRIMARY **DigiKey** 1528-2748-ND 321 in stock*
Adafruit ALS-PT19 analog light sensor breakout
[digikey.com/en/products/detail/adafruit-industries-llc/2748/5775538](https://www.digikey.com/en/products/detail/adafruit-industries-llc/2748/5775538)

\$7.50
3 × \$2.50

BACKUP **Adafruit** 2748 In stock
Adafruit ALS-PT19 Analog Light Sensor Breakout
adafruit.com/product/2748

\$7.50
3 × \$2.50

2.5 to 5.5 V, analog output. Cheaper backup: Adafruit 161 photocell (\$0.95, needs a 10k resistor).

08 Red LED, 5 mm

Qty 10

Mosquito eyes that flash on a hit (2 per mosquito, plus spares).

PRIMARY **DigiKey** 754-1264-ND 145,526 in stock*
Kingbright WP7113ID red diffused 5 mm
[digikey.com/en/products/detail/kingbright/WP7113ID/1747663](https://www.digikey.com/en/products/detail/kingbright/WP7113ID/1747663)

\$1.45
10 × \$0.14

BACKUP **Adafruit** 297 In stock
Super Bright Red 5mm LED (25 pack)
adafruit.com/product/297

\$8.00

Price is the 10-piece rate.

09 220 ohm resistor

Qty 10

One per LED.

PRIMARY **DigiKey** 220QBK-ND 77,901 in stock*
Yageo CFR-25JB-52-220R, 1/4 W
[digikey.com/en/products/detail/yageo/CFR-25JB-52-220R/1295](https://www.digikey.com/en/products/detail/yageo/CFR-25JB-52-220R/1295)

\$0.36
10 × \$0.036

BACKUP **Adafruit** 2780 In stock
Through-Hole Resistors, 220 ohm (25 pack)
adafruit.com/product/2780

\$0.75

Price is the 10-piece rate.

10 Thin 4-wire cable, 1 m

Qty 1

Runs down the fishing line to the sensor and eyes: power, ground, sensor, LEDs.

PRIMARY **DigiKey** 1528-2694-ND 16 in stock*
Adafruit 3891 silicone ribbon cable, 4 wire, 28 AWG, 1 m
[digikey.com/en/products/detail/adafruit-industries-llc/3891/9603617](https://www.digikey.com/en/products/detail/adafruit-industries-llc/3891/9603617)

\$1.95

BACKUP **Adafruit** 3890 In stock
Silicone ribbon cable, 10 wire, 28 AWG, 1 m (peel off 4)
adafruit.com/product/3890

\$3.95

Low stock at DigiKey: order early.

Sound

11 Speaker module

Qty 1

Laser pew, cannon boom, lock-on beep, victory sound. Volume knob.

PRIMARY	DFRobot FIT0449 In stock	Gravity: Digital Speaker Module	\$6.00
dfrobot.com/product-1401.html			
BACKUP	Adafruit 2130 + 3923 In stock	PAM8302 2.5 W mono amp (\$3.95) + mini oval 8 ohm speaker (\$1.95)	\$5.90
adafruit.com/product/2130			

Plugs into the Xia mi P0 port (5 V); MakeCode plays sound on P0. Backup speaker page: adafruit.com/product/3923 (solder the wires to the amp).

Power

12 Turret battery pack, 4xAA

Qty 1

External power for the Xia mi board: servos, camera, speaker, laser.

PRIMARY	DFRobot FIT0918 In stock	4xAA battery holder with power switch, DC 2.1 jack, cover	\$2.00
dfrobot.com/product-2649.html			
BACKUP	Adafruit 771 In stock	3xAA battery holder with on/off switch (4.5 V)	\$3.95
adafruit.com/product/771			

About 6 V, inside the Xia mi's 5 to 12 V range. The 3xAA backup matches the micro:Driver (3.5 to 5.5 V). A 5 V USB power bank into the board's USB port also works.

13 micro:bit battery pack

Qty 2

Powers the wand micro:bit, and the turret micro:bit if the board does not.

PRIMARY	DFRobot FIT0625 In stock	2xAAA battery holder with cover and switch, PH2.0	\$2.50
dfrobot.com/product-1855.html			
BACKUP	Adafruit 4193 In stock	2xAA battery holder with on/off switch and JST PH	\$3.90
adafruit.com/product/4193			

The micro:bit battery socket is JST PH 2.0 mm. Adafruit's 2xAAA version (4191) is out of stock, so the backup is 2xAA.

Wiring and hardware

14 M2 screw, nut, and standoff kit

Qty 1

Every printed part: base, turret head, wand handle, sensor pod, and backdrop parts.

PRIMARY	DFRobot FIT0665 In stock	Metal screw, nut and mounting kit, M2, 320 pcs	\$12.90
dfrobot.com/product-2023.html			
BACKUP	DigiKey 1738-FIT0665-ND 321 in stock*	DFRobot FIT0665 (same kit)	\$12.90
digikey.com/en/products/detail/dfrobot/FIT0665/12324933			

Prints use M2 clearance holes (2.2 mm) and M2 nut traps. Exceptions: HuskyLens uses its own M3 kit; servos use their own horn screws.

15 Gravity 3-pin cables (10 pack)

Qty 1

Servo, speaker, and laser leads to the board, plus spares.

PRIMARY	DFRobot	FIT0031	In stock	
	Gravity: Analog Sensor Cable, 30 cm (10 pack)			\$6.00
	dfrobot.com/product-128.html			
BACKUP	DigiKey	1738-1031-ND	111 in stock*	
	DFRobot FIT0031 (same cables)			\$6.00
	digikey.com/en/products/detail/dfrobot/FIT0031/6588453			

Order sheets

DFRobot cart (primary)

#	SKU	PART	QTY	PRICE
01	MBT0039	micro:bit V2	2	\$45.80
02	MBT0042	Expansion board (turret)	1	\$29.90
03	SEN0305	HuskyLens AI camera	1	\$34.90
04	SER0049	Micro servo	3	\$14.70
11	FIT0449	Speaker module	1	\$6.00
12	FIT0918	Turret battery pack, 4xAA	1	\$2.00
13	FIT0625	micro:bit battery pack	2	\$2.50
14	FIT0665	M2 screw, nut, and standoff kit	1	\$12.90
15	FIT0031	Gravity 3-pin cables (10 pack)	1	\$6.00
Subtotal				\$154.70

DFRobot has offered free US shipping over \$50 as a limited-time deal; check at checkout.

DigiKey cart (primary)

#	DIGIKEY / PART	PART	QTY	PRICE
05	VLM-650-03-LPT-ND	Class 2 laser module, under 1 mW	1	\$25.03
06	Adafruit 805	Laser arm switch	1	\$0.95
07	1528-2748-ND	Light sensor	3	\$7.50
08	754-1264-ND	Red LED, 5 mm	10	\$1.45
09	220QBK-ND	220 ohm resistor	10	\$0.36
10	1528-2694-ND	Thin 4-wire cable, 1 m	1	\$1.95
Subtotal				\$37.24

Resistors and LEDs are sold loose; type the quantity in the cart.

From local stores or already on hand

ITEM	AMOUNT	WHERE
AA batteries	4 plus 4 spares	Grocery store
AAA batteries	4 plus 4 spares	Grocery store
Stuffed mosquito toy (the target)	1	Toy store or online
Short fishing rod or 3/8" dowel, clear 10 lb line	1	Sporting goods or hardware store
Black foam board, 20x30 in, 3/16" thick	1	Craft store
1/4" wood dowel	1	Craft or hardware store
Small zip ties and hook-and-loop tape	1 pack each	Hardware store
PLA filament, any color plus a little white	About 310 g + 11 g white	On hand (check for white)
Solder, heat-shrink tubing, hot glue	Small amounts	On hand
Micro-USB data cables	2	Probably on hand
Paper data sheets, printed	15 per test	Print at home

What the design locks in

- **Design A:** pan and tilt camera turret, HuskyLens on top, laser below it, boresighted.
- **Printed tilt stop** plus a code limit: cannot aim above the backdrop.
- **Laser path:** 5 V port, through the arm switch, through the board relay, to the laser.
- **Ports:** HuskyLens on the 5 V HuskyLens port; pan servo P1, tilt servo P2, speaker P0 (all 5 V); P8 spare.
- **Wand:** second micro:bit in a printed handle, 2xAAA pack, 4-wire cable down the line to the mosquito (sensor + LED eyes).
- **Fasteners:** M2 everywhere in printed parts; M3 only for the HuskyLens.

If the primary board gives trouble

- Switch to the **micro:Driver** (backup #02): servos on its servo headers, laser on motor channel M1, power from the 3xAA backup pack.
- Its I2C port voltage is not documented. Test the HuskyLens on it before relying on it; if it browns out, power the HuskyLens from a 5 V USB power bank.
- The printed base has room for either board (Xia mi 57 × 87 mm is the larger).